In-class Lab-Assignment 7

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CYSE1007: Cloud Security

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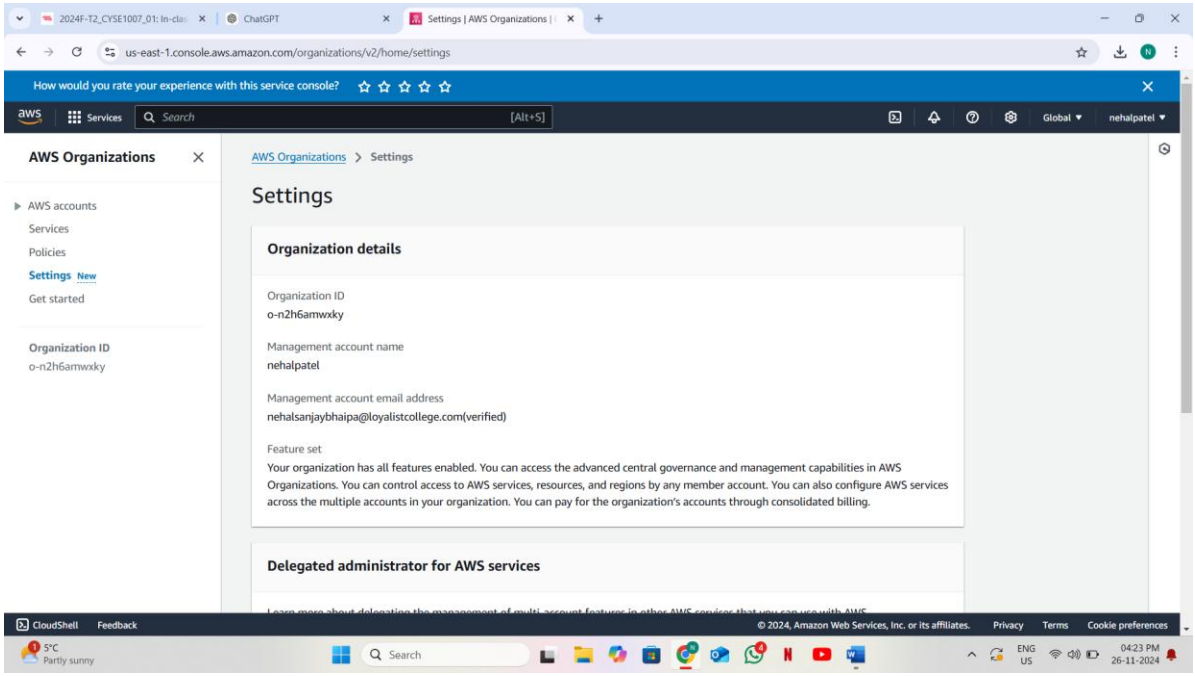
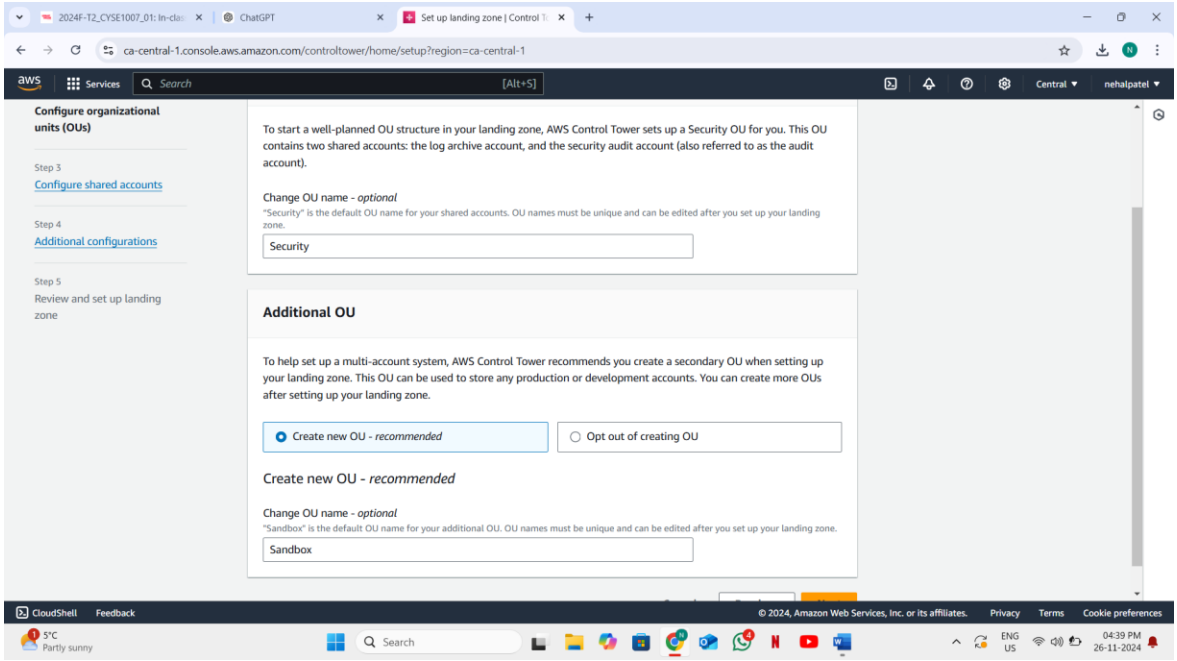
Step 1: Enable AWS Organizations

1. Log in to your AWS Management Console.
2. Navigate to **AWS Organizations** and create an organization:
   - Select **Enable all features** to allow full governance by Control Tower.
   - 

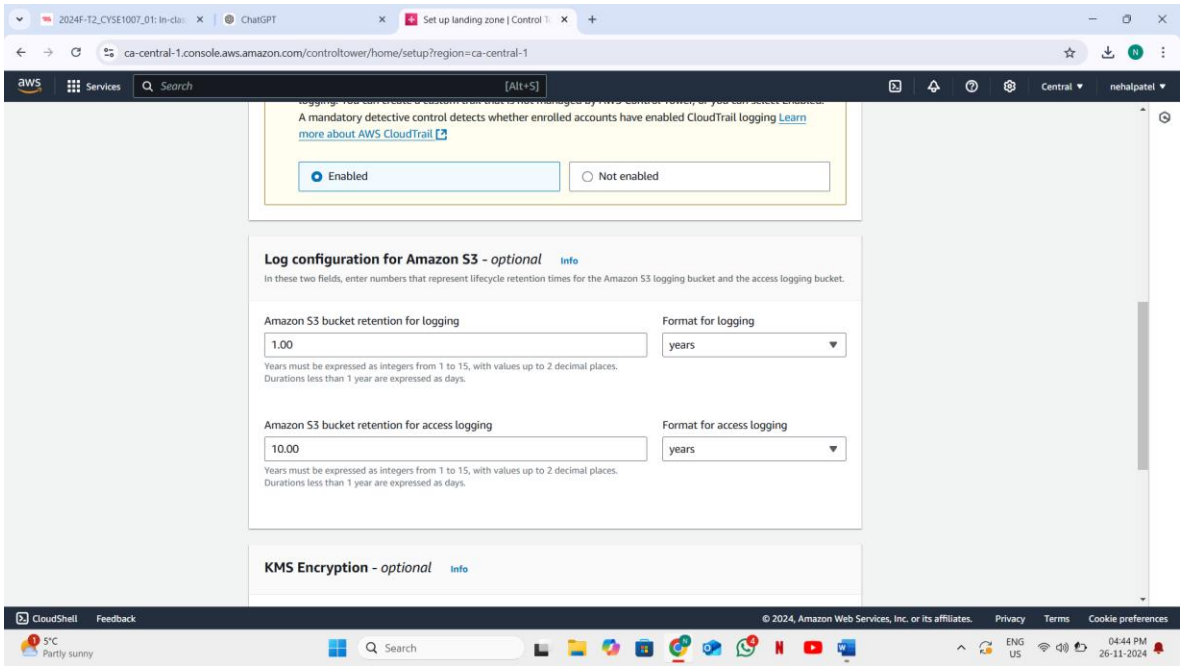
3. Note the **Management Account** (root account), which will be used to set up Control Tower.

Step 2: Set Up AWS Control Tower

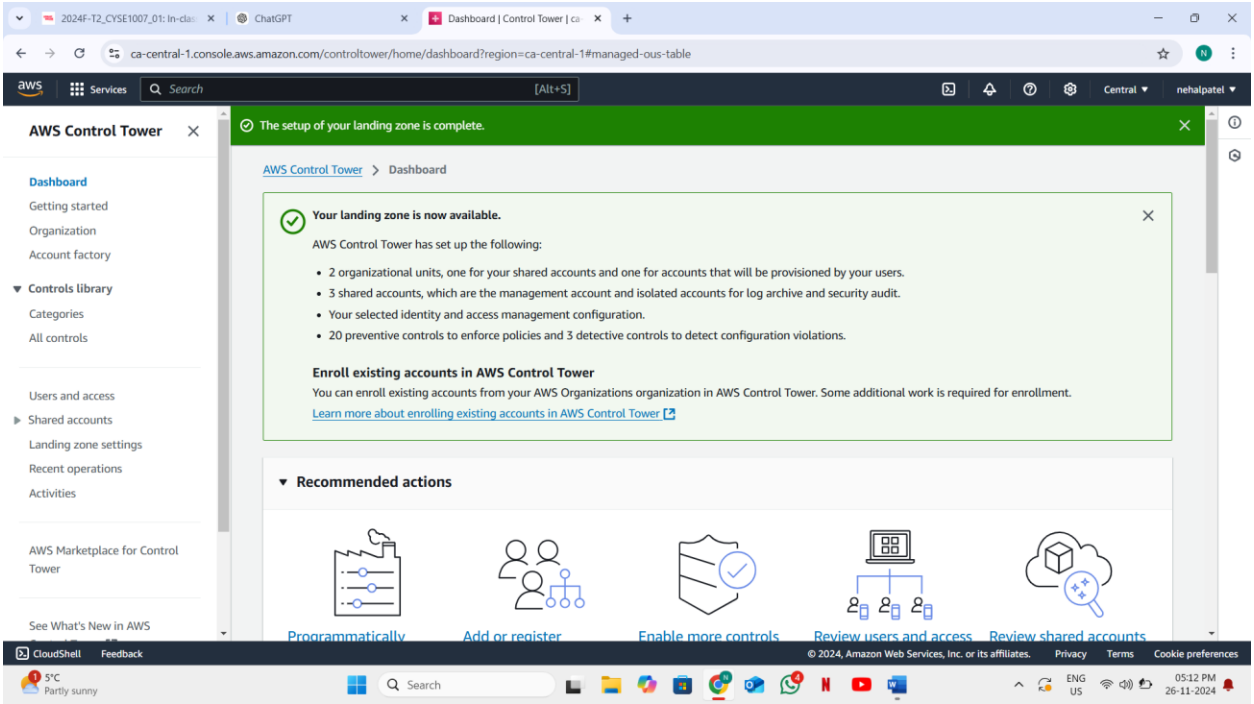
1. Navigate to **AWS Control Tower**.
2. Click **Set up landing zone**.
3. Review and confirm the prerequisites, such as enabling AWS Config and CloudTrail in the region.
4. Configure the landing zone setup:
   - Choose the management account's email address.
   - Specify organizational unit (OU) names, e.g., Core and Sandbox.



- Define an S3 bucket for logs and centralized logging services.



5. Start the deployment process (this can take 30–60 minutes).



Step 3: Create Additional Accounts

1. Navigate to the **AWS Control Tower dashboard**.
2. Use the **Account Factory** feature to create accounts:
    - Create a new account (e.g., StudentTestAccount) and assign it to the Sandbox OU.
    - Ensure the account is provisioned successfully.

    Error: AWS Control Tower can't create your account due to potential drift in your landing zone. Check your landing zone and try using the advanced account provisioning method to create your account.

    How I tried to solve this problem:

    Steps to Resolve Landing Zone and Account Issues:

    1. Verify Landing Zone Health

- Check Status:
    - Go to AWS Control Tower > Landing zone settings.
    - Review warnings or errors under Landing zone drift detection or Setup progress.
- Detect Drift:
    - If drift is detected, select Detect drift to review the issues.

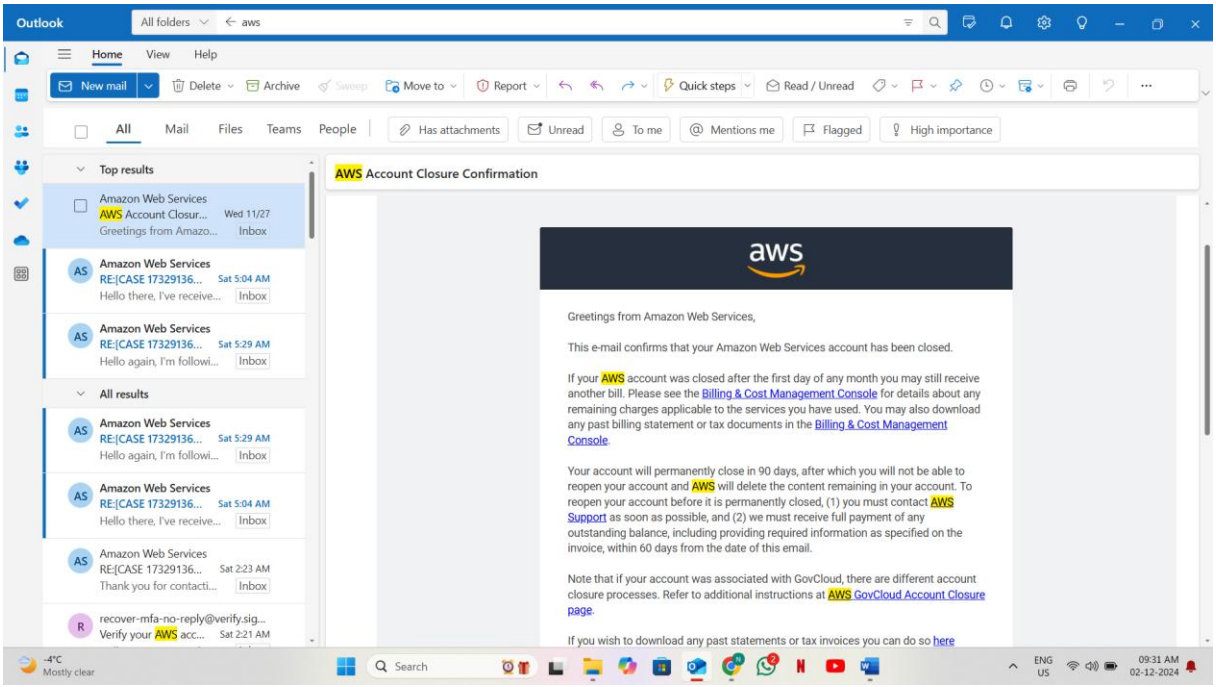
- Resolve Issues:
    - Restore missing resources (e.g., S3 buckets, IAM roles, CloudTrail trails).
    - Reapply required permissions for AWS Control Tower.

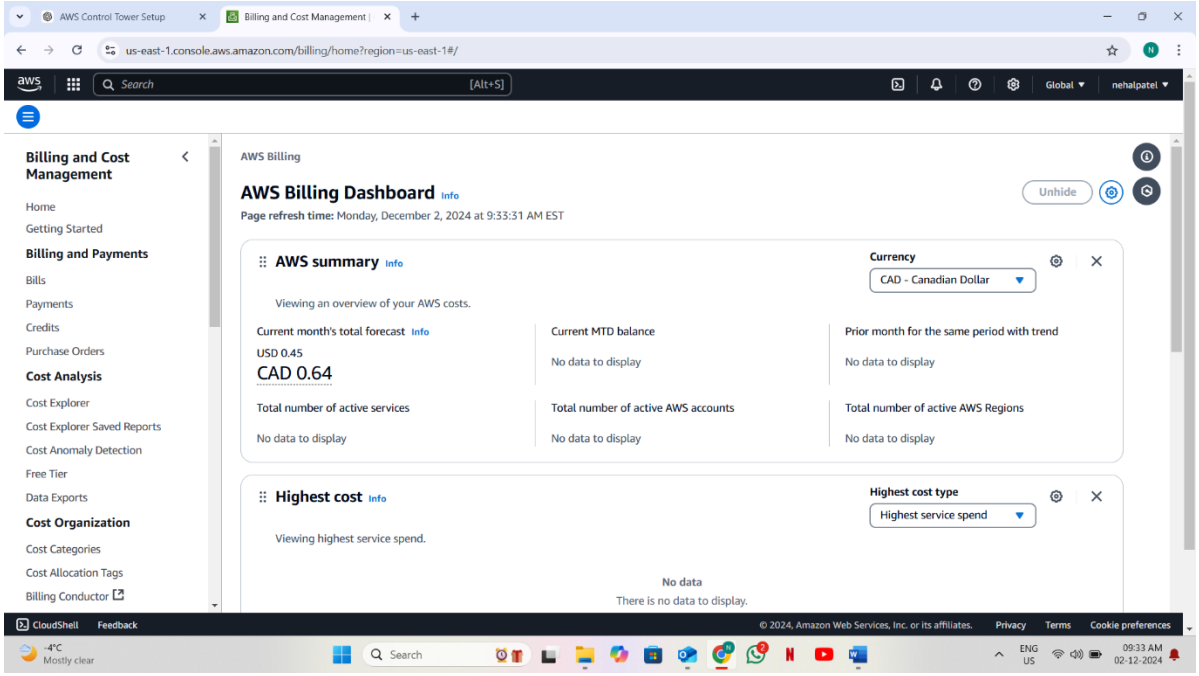
2. Update the Landing Zone

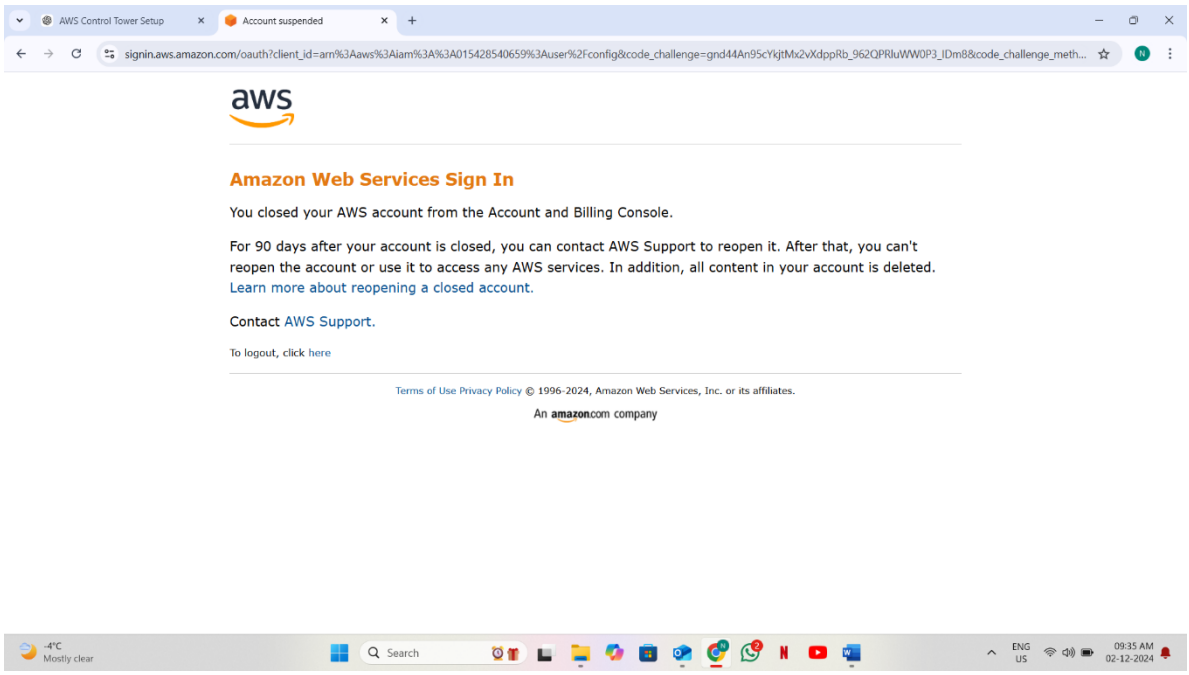
- Go to Control Tower dashboard > Landing zone settings.
- Click Update landing zone to reapply configurations and resolve drift.
- Ensure all required AWS services (AWS Organizations, IAM Identity Center, CloudTrail) are active.

Challenges Encountered

Despite following all the recommended steps to resolve the issues with AWS Control Tower, I was unable to proceed beyond Step 3. After applying the steps to detect and resolve drift, update the landing zone, and attempt advanced account provisioning, I encountered persistent issues that prevented further progress. Additionally, I noticed **unexpected charges** on my account despite deleting everything I created for this assignment. The charges increased the next day when I attempted to access the billing page. As a result, I had to **suspend my AWS account and block the card** associated with it to prevent further deductions. Unfortunately, this situation left me unable to complete the remaining steps for this assignment. I have attached a screenshot to provide context for the issue encountered. I sincerely apologize for being unable to continue further and submit a fully resolved assignment.







Step 4: Implement and Test Guardrails

1. Navigate to **Guardrails** in the Control Tower console.

2. Enable **mandatory guardrails** like:
    - Prevent public access to S3 buckets.
    - Require multi-factor authentication (MFA) for root accounts.
3. Enable **optional guardrails** such as restrictions on regions where resources can be launched.

Testing Guardrails:

- Log in to the StudentTestAccount.
- Attempt to create an S3 bucket with public access or launch resources in a restricted region.
- Observe compliance and non-compliance messages in the AWS Control Tower dashboard.

Step 5: Verify Centralized Logging

1. Navigate to the S3 bucket configured for centralized logging.
2. Review logs generated by:
    - **AWS Config**: Tracks resource changes.
    - **CloudTrail**: Logs account activity.
    - **Control Tower**: Captures guardrail violations.
3. Demonstrate how these logs help monitor and troubleshoot account activity.

Use Cases for AWS Control Tower

1. **Multi-Account Management**:
    - Automatically apply governance across multiple accounts in the organization.
2. **Compliance and Security**:
    - Use guardrails to enforce organizational policies, such as restricting regions or securing S3 buckets.
3. **Centralized Logging**:
    - Consolidate activity logs to a single S3 bucket for monitoring and auditing purposes.
4. **Simplified Onboarding**:

- Create pre-configured accounts quickly using Account Factory.

Deliverables for Students

1. Screenshot of the Control Tower landing zone and organizational units.
2. Log files showing activities and compliance violations.
3. Documentation of guardrails enabled and their observed effects.

Cleanup Instructions

1. Delete all accounts created using Account Factory.
2. Disable Control Tower and delete the landing zone.
3. Delete resources like S3 buckets and logs generated.
4. Remove the organization to avoid ongoing charges.